

Introduction

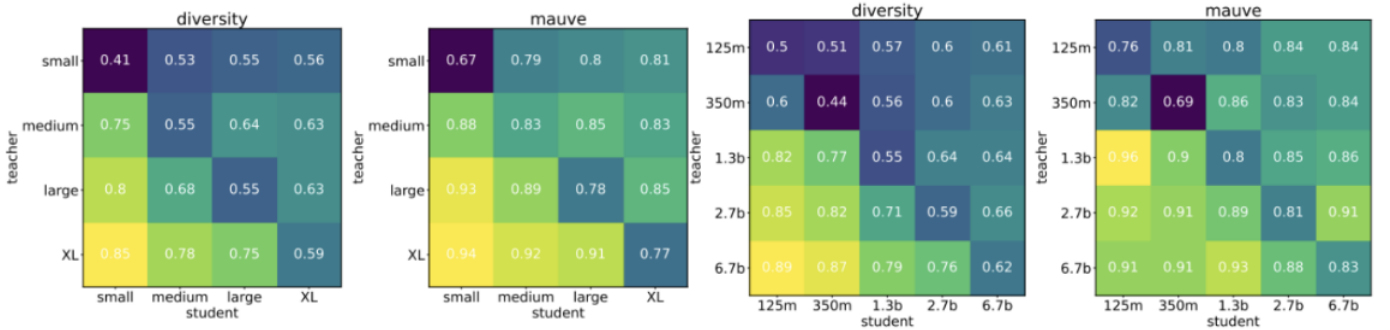
Text generation using existing methods such as greedy decoding and nucleus sampling suffers from topic drift and repetitiveness. In the paper, *Contrastive Decoding: Open-ended Text Generation as Optimization*, Xiang Lisa Li et al. present a new decoding method that addresses these issues by taking advantage of the strengths and weaknesses of existing models of different sizes. In particular, they improve the performance of the strong model by penalizing the outputs that the weak model prefers, essentially “subtracting” weaknesses like repetition and topic drift from the strong model. This is an efficient solution because it uses existing models, avoiding any additional fine-tuning or training.

Chosen Result

Experiment 1 compares Contrastive Decoding (CD) to other sampling methods, which is the main objective of the paper using this equation $\log P_{expert}(x) - \log P_{amateur}(x)$. Paper results:

	name	wikinews			wikitext			story		
		DIV	MAUVE	COH	DIV	MAUVE	COH	DIV	MAUVE	COH
OPT-13B	max prob	0.08	0.3	0.65	0.03	0.08	0.63	0.02	0.05	0.51
	k=50	0.91	0.92	0.64	0.72	0.77	0.64	0.91	0.9	0.51
	p=0.95	0.92	0.92	0.62	0.92	0.89	0.55	0.93	0.91	0.48
	typical=0.95	0.94	0.9	0.59	0.89	0.86	0.58	0.95	0.91	0.46
	CS(Su et al., 2022)	0.92	0.87	0.59	0.87	0.77	0.52	0.81	0.78	0.47
	CD	0.94	0.94	0.69	0.91	0.91	0.69	0.89	0.94	0.62
GPT2-XL	max prob	0.04	0.14	0.65	0.02	0.05	0.62	0.01	0.03	0.49
	k=50	0.92	0.88	0.64	0.87	0.79	0.61	0.91	0.87	0.51
	p=0.95	0.94	0.9	0.6	0.92	0.87	0.57	0.94	0.91	0.46
	typical=0.95	0.95	0.91	0.56	0.95	0.84	0.53	0.96	0.88	0.43
	CS(Su et al., 2022)	0.93	0.82	0.62	0.86	0.75	0.59	0.88	0.78	0.48
	CD	0.92	0.94	0.69	0.89	0.92	0.69	0.83	0.94	0.64

Experiment 2 studies the effect of relative model size on CD performance and helps determine model sizes for Experiment 1. Here are the paper results:



Experiment 3 is our own attempt to test the diversity-coherence trade-off and validate the paper’s $V_{\text{head}}(x_{<i}) = \{x_i \in V : p_{\text{EXP}}(x_i | x_{<i}) \geq \alpha \max_w p_{\text{EXP}}(w | x_{<i})\}$ plausibility constraint. We confirmed $\alpha = 0.1$ as the optimal threshold for balancing the 3 metrics.

Methodology

The re-implementation approach uses a Contrastive Decoding (CD) objective that maximizes the log-probability difference between a large expert model, such as GPT-2 XL or OPT-6.7B, and a smaller amateur model, such as GPT-2 Small or OPT-125M, while enforcing an adaptive plausibility constraint to ensure fluency. Evaluation is conducted by generating 256-token continuations from

32-word prompts across 100 samples from the Wikitext-103, CC-News, and PG-19 datasets. Generation quality is measured using Diversity (DIV), MAUVE, and Coherence (COH), which is calculated using SimCSE sentence embeddings. To maintain hardware feasibility on limited resources, the codebase implements 8-bit quantization and fp16 precision for the larger expert models. The methodology uses both greedy and beam search variants with beam size 5, and also includes a sensitivity sweep for the plausibility threshold α ranging from 0.01 to 0.2, alongside tuned amateur temperatures τ .

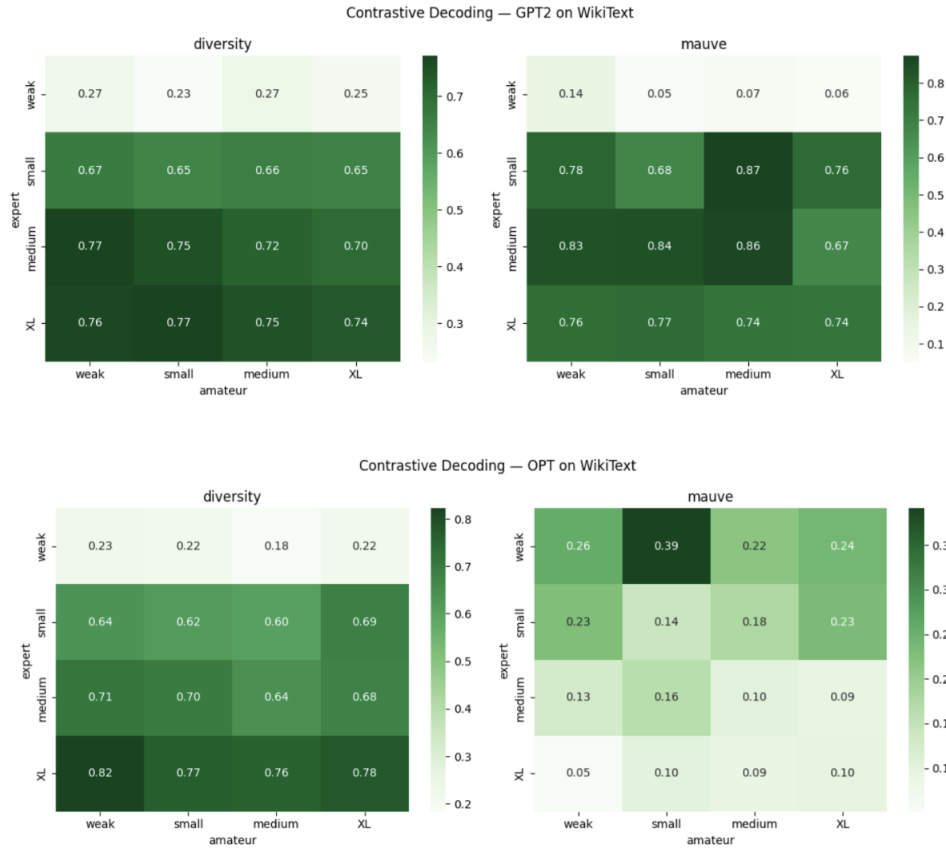
Results and Analysis

For experiment 1, we compared CD against several baseline sampling methods, including greedy decoding, top-50 sampling, and nucleus sampling(w/ p=0.95). We ran all three datasets across five sampling methods and evaluated the outputs using MAUVE, DIV, and COH metrics. For CD, we used the same alpha and temperature values as the original paper. The paper used full CD with beam search, but this was expensive to run, so we mainly used a CD greedy, which only keeps the maximum scoring option instead of keeping the top 5 candidates. We also ran 10 iterations of CD beam search as a smaller-scale test. In addition, we completed a full run using GPT-2 models as well as larger OPT models, and found consistent results across these setups. Overall, CD greedy performed about on par with the best of the other sampling methods, while CD beam search outperformed every other method in the smaller test, except for MAUVE, which was inconsistent due to the low number of iterations. This supports the paper’s main claim that contrastive decoding can improve generation quality by comparing strong and weak model predictions. The main challenge we encountered was computational cost. These experiments took a long time, and interruptions during the day often stopped the runs, so most testing had to be done overnight.

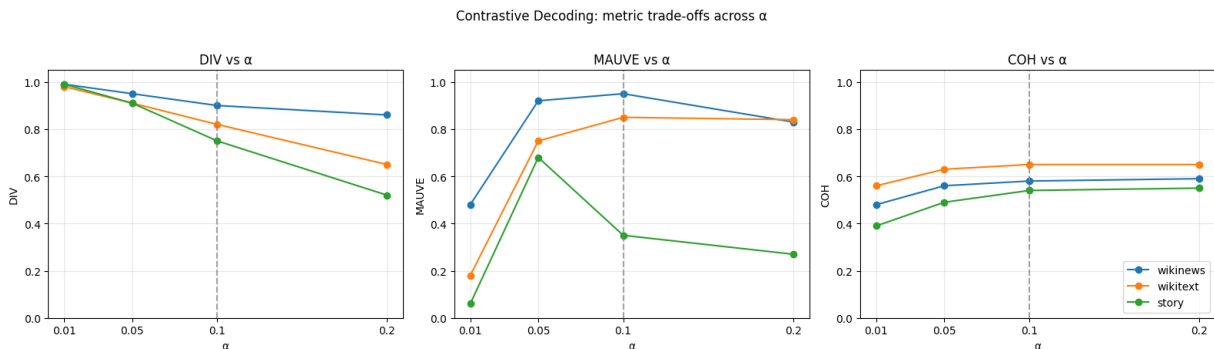
	name	wikinews			wikitext			story		
		DIV	MAUVE	COH	DIV	MAUVE	COH	DIV	MAUVE	COH
GPT2-XL	max prob	0.17	0.15	0.58	0.09	0.10	0.48	0.10	0.12	0.47
	k=50	0.91	0.91	0.63	0.89	0.84	0.59	0.90	0.97	0.51
	p=0.95	0.94	0.93	0.58	0.93	0.90	0.54	0.94	0.98	0.45
	CD	0.90	0.95	0.66	0.82	0.85	0.65	0.89	0.95	0.58
	CD Beam	0.96	0.67	0.66	0.96	0.76	0.65	0.96	1.00	0.65
OPT	max prob	0.30	0.37	0.57	0.07	0.02	0.58	0.04	0.02	0.46
	k=50	0.89	0.85	0.62	0.89	0.54	0.61	0.90	0.81	0.52
	p=0.95	0.91	0.89	0.60	0.92	0.67	0.58	0.90	0.67	0.59
	CD	0.90	0.75	0.63	0.88	0.80	0.66	0.85	0.72	0.60
	CD Beam	0.91	0.98	0.69	0.91	0.82	0.70	0.92	0.82	0.63

In Experiment 2, we confirmed that the highest CD scores are achieved where the gap between the expert and amateur models is the largest. This supports the paper’s choice to evaluate CD run on a small amateur and large expert model and their claim that larger size gaps between the expert and amateur models result in higher quality text generation. While our overall trend matched, the magnitudes of our results were lower than the paper’s across the board, with significant discrepancies in MAUVE. We attribute the magnitude difference to our substitution of CD-Beam for CD-greedy. Additionally, MAUVE tends to fluctuate and is sensitive to the sample size it’s

calculated on. Given more compute, we hypothesize that increasing our sample size significantly would stabilize MAUVE.



Our Experiment 3 used the GPT-2 model to find the optimal plausibility threshold α . As expected, DIV showed an inverse relationship with α , crashing at $\alpha = 0.2$ as the model became restricted to repetitive high-probability tokens. COH rose sharply from low α values and plateaued around $\alpha = 0.1$, validating its role in filtering out nonsensical false positives. While our MAUVE scores for news and Wikipedia peaked at the paper’s recommended $\alpha = 0.1$, the story dataset peaked earlier, around $\alpha = 0.05$, and showed lower overall similarity to human text at higher thresholds. We attribute this discrepancy to challenges with our smaller evaluation sample of 100 prompts and the substitution of PG-19 for the original BookCorpus. Despite these small variations, the overall results strongly support the paper’s idea that the adaptive plausibility constraint is essential for balancing creativity and topic stability.



Reflections

This project showed us that re-implementing a paper is much harder than it looks, even when the method does not require training a new model. At first, our results did not match the paper, and we learned that many small implementation details can have a large effect. Some of the original datasets, such as BookCorpus, were no longer available, so we had to find duplicates to match the original setup. One issue was that our version of the story dataset did not clearly mark where one story ended and another began. This meant that some prompts may have contained parts of two different stories, which confused the model. Another important lesson was that metrics must be implemented exactly the same way to get similar results. For example, the paper defines diversity using 2-grams to 4-grams, but we originally included 1-grams, which made our diversity scores much lower. Fixing these issues was time-consuming because each full experiment took around 8–10 hours to rerun, and we ran multiple experiments across several datasets and decoding methods.

We also learned that compute cost and randomness are major parts of replication work. Even though contrastive decoding does not require model training, running the experiments still used a large amount of time, around 500 Google Colab compute units. At first, we used A100 GPUs because the models did not fit on T4s, but we later found that L4 GPUs could fit the models while using much fewer compute units, which helped reduce the cost. Another takeaway was that results can change noticeably between runs, especially for metrics like MAUVE, so a single run may not be enough to draw strong conclusions. Overall, this project taught us that reproducing research results requires careful data preparation, exact metric implementation, efficient use of compute, and patience when results do not match immediately.

For future work, we would like to make our implementation closer to the original paper by using CD beam search instead of CD greedy. We used greedy decoding because it was much faster(20x faster), but the paper used beam search, and this difference may explain some gaps in performance. Another useful extension would be to include human evaluation, because automatic metrics like diversity, COH, and MAUVE do not always fully capture whether the generated text feels better to a person. Finally, it would be interesting to test contrastive decoding on other tasks, such as summarization, dialogue, and question answering, to see whether the method is useful beyond open-ended text generation.

References

- [1] Xiang Lisa Li, Ari Holtzman, Daniel Fried, Percy Liang, Jason Eisner, Tatsunori Hashimoto, Luke Zettlemoyer, and Mike Lewis. Contrastive decoding: Open-ended text generation as optimization, 2023.
- [2] Richa Gadgil. Combining large and small llms to boost inference time and quality. <https://towardsdatascience.com/combining-large-and-small-llms-for-inference-time-and-quality-boosts-1779b6b5100b/>, December 2024. Towards Data Science.
- [3] OpenAI and Anthropic. Chatgpt and claude. Large language models used for understanding the paper, code inspiration, bug fixes, and optimizations.